The Causal Algorithm at Work

Session VII · 19 October 2026 — the opioid epidemic as a worked example

Class notes · an automated summary of the live session, structured for study. Where these notes and the assigned papers disagree, the papers win.

Concept

- The session turned the opioid epidemic into a worked exercise in causal inference: whether marketing caused overdose deaths, judged by a structured algorithm rather than a headline
- The tools: naming exposure and outcome precisely, checking the order in time, naming the rivals (chance, bias, confounding, reverse causation), and testing each observation against them

The opioid timeline used in class

- 1980 — the Porter & Jick letter in the NEJM: addiction reported rare in inpatients given narcotics; the journal's authority carried the claim far beyond its five sentences
- 1996 — OxyContin launches, marketed as safe for outpatients; ~300,000 prescriptions in the first year
- 2001 — prescriptions around the first large peak (see the check-block on this figure)
- 2007 — the manufacturer pleads to felony misbranding; ~$600 million in penalties — and prescribing keeps rising
- 2010–2012 — prescription rate plateaus around 81 per 100 people
- 2013 — Open Payments makes industry payments to doctors public; the landmark studies become possible only after this
- 2015 — prescriptions down ~13%
- 2017 — ~70,000 overdose deaths, ~47,000 opioid-related; synthetic-opioid deaths up ~45% in one year
- 2017–2018 — the two JAMA-family studies linking marketing payments to prescribing and mortality

One question becomes three

1. Did marketing make doctors prescribe more? Probably yes — when payments and campaigns were restricted, prescribing volume dropped; the cessation-of-exposure logic points the same way as the payment data

2. Did the campaign kill people in 1996–2010? Yes, with caveats — the triplicate-prescription states (California, Idaho, Illinois, New York, Texas) required a three-copy form for Schedule II drugs; OxyContin marketing avoided them, and their overdose growth stayed lower after 1996. A natural experiment: harder prescribing, fewer deaths. The caveat is the targeting effect: marketing went where prescribing culture already lived, so some of the geography predates the campaign

3. Is marketing driving deaths from 2017 on? No — prescriptions had already fallen while deaths kept climbing; the driver is illicit fentanyl, an independent epidemic by then

The four-step algorithm

Step 1 — name X and Y precisely. How you name them shapes the answer; one article usually holds several X–Y pairs that must be judged separately

Step 2 — can the design show the order in time? Cross-sectional designs cannot: everything is measured at once. A vaccine blamed for an infection acquired before the shot is the everyday version; non-live vaccines cannot cause the infection at all, unlike live attenuated ones which can

produce mild symptoms

Step 3 — are there rivals? Chance; bias in how outcomes were counted; a confounder (poverty in the 19th-century cholera studies — associated with both the water and the deaths, but not the cause); reverse causation — the outcome attracting the exposure, as when marketing targets places that already prescribe heavily

Step 4 — test each observation against each rival: could the rival alone produce what the study saw? If yes, the observation is decoration, not evidence

Methodological asides from the session

- Surrogate outcomes: distribution volume standing in for overdose risk when the true outcome is hard to measure
- Data availability shapes science: the causal studies waited for Open Payments (2013) — what cannot be measured cannot be studied
- Reputation as a causal shortcut: the 1980 letter's authority came from the journal, not the evidence — a five-sentence letter steered a prescribing era
- The Netherlands abortion example: regional religiosity confounded both the type of abortion recorded and the willingness to report — medical studies are social studies
- A student (Konstantin) raised the unmeasured pathway: the launch may have seeded recreational use far beyond prescriptions — plausible, and invisible to official data

Might show up on the control

- The four steps, applied to a new paper: name X and Y; can this design see time order; list the rivals; test each against the observations
- Why cross-sectional designs cannot establish order
- Cessation of exposure as supporting evidence, and its limits
- Two stock examples of reverse causation: the pre-vaccination infection; the cancer that explains the weight loss "before" the diagnosis
- Confounder vs reverse causation: a third factor versus an inverted arrow

Check before you lean on it

- Triplicate states: the three-copy programmes predate OxyContin by decades — California's dates from 1939 — and were prescription-control measures, not a response to pre-1996 overdose rates. What the natural experiment uses is only this: the barrier existed where the campaign could not bite. The transcript's "they had higher overdose rates before 1996, which is why they introduced the rules" reverses the history.
- The "2001 ≈ 6 million prescriptions" line conflicts with the session-VI notes ("prescribing peaked ~2010"); OxyContin-specific prescriptions and all-opioid prescribing are different curves. Check before citing either number.
- The 2017–18 studies are Hadland et al., JAMA Internal Medicine 2018 and JAMA Network Open 2019 (the transcript said NEJM).